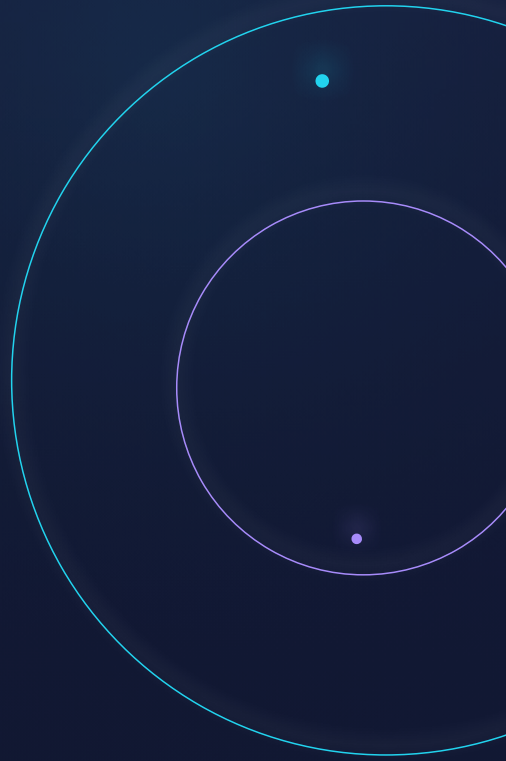
HACKATHON PITCH · QUANTUM OPTIMIZATION FOR EV CHARGING STATIONS

Q·Grid

Every EV becomes an agent. The Center does the thinking.

A software-only quantum-optimized charging orchestrator — piloted on Bengaluru traffic data.



Q

by JustCode

— in collaboration with Qub3RAM.team

15 agents · 5 stations · 344-binary QUBO

THE PROBLEM

Charging today is first-come, first-served — and everyone loses.

Drivers queue at the nearest station without knowing where capacity is free. Stations fill in lopsided ways — one port oversubscribed while another sits idle. And charging demand lands squarely on the evening peak, when grid electricity costs 4.5x the midday solar price.

4.5x

evening tariff vs midday solar

(₹18.00 vs ₹4.00 /kWh in our Bengaluru window)

0

coordination between vehicles today

no agent, no forecast, no global view

21%

of waiting cost the FCFS baseline wastes in our own simulation

(₹1,830 of ₹4,206 total)

The constraint: we cannot build new stations. We can only decide better who plugs in where, and when.

THE IDEA

A tiny agent in every EV. One global brain at the Center.

On every vehicle

Q-Agent SDK — pre-installed by OEMs. Reports GPS position, state of charge and a pseudonymous ID every 30 seconds.

No cameras, no PII, no new hardware.

At the Center

The Center ingests the whole fleet, triages urgency with AI, prunes to the 2 nearest stations per vehicle, and encodes the global assignment as one QUBO.

Solvers return a city-wide schedule: who charges where, starting when, at what price.

DRIVERS NEED NOTHING — no app, no dashboard, no login. Charging is a background task Q Grid performs; the schedule arrives over the air.

01 **SENSE** telemetry in → 02 **OPTIMIZE** quantum + classical + AI → 03 **ACT** schedule out to every agent

Pure software. Works with any OEM that ships the SDK — the open SDK is the moat, not the hardware.

SYSTEM ARCHITECTURE

Five layers, from telemetry to schedule — all software.



L1 EV Agent Layer

Q-Agent SDK in every vehicle

L2 Connectivity & Privacy Gateway

pseudonymization, regional sharding

L3 THE CENTER

`triage → prune → QUBO → 4 solvers → feasibility guard`

L4 Charging Network

5 Bengaluru stations, time-of-use tariff

L5 Output & Feedback

schedule push, dashboards, measured impact

THE MATH

One QUBO holds the whole city-wide decision.

01 Variables

`y[v,s,τ]` = 1 iff vehicle `v` starts one continuous charging session at station `s` in half-hour slot `τ`. `344 binaries` in the demo.

02 Objective (linear, ₹)

```
energy cost E(v,s)·tariff(τ) + travel 9 ₹/km · dist(v,s) + waiting
w(v)·(τ − arr_v),
```

where w = ₹260 NOW / ₹150 SOON / ₹80 CAN-WAIT per slot.

03 Constraints

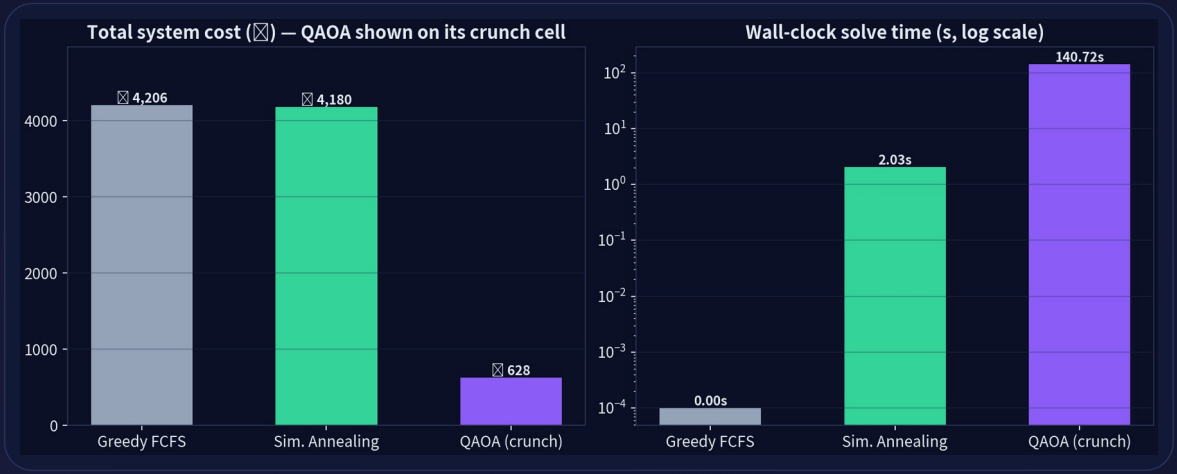
each vehicle charges exactly once (penalty λ_1=4000); port capacity $\Sigma x \le$ ports encoded exactly with slack variables `u` (weights `m` = 1..ports) so empty ports are never punished.

✦ Why this encoding wins

- ✓ **Continuous sessions guaranteed by construction** — no split charging
- ✓ **Exact capacity inequality** — SA stopped overfilling stations (the bug that once made greedy look smart)
- ✓ **AI-set urgency weights** make fairness a number, not an afterthought
- ✓ **Scales with contested slots, not fleet size** — the quantum entry point

EXPERIMENT

We raced four solvers on the same QUBO — the baseline loses.



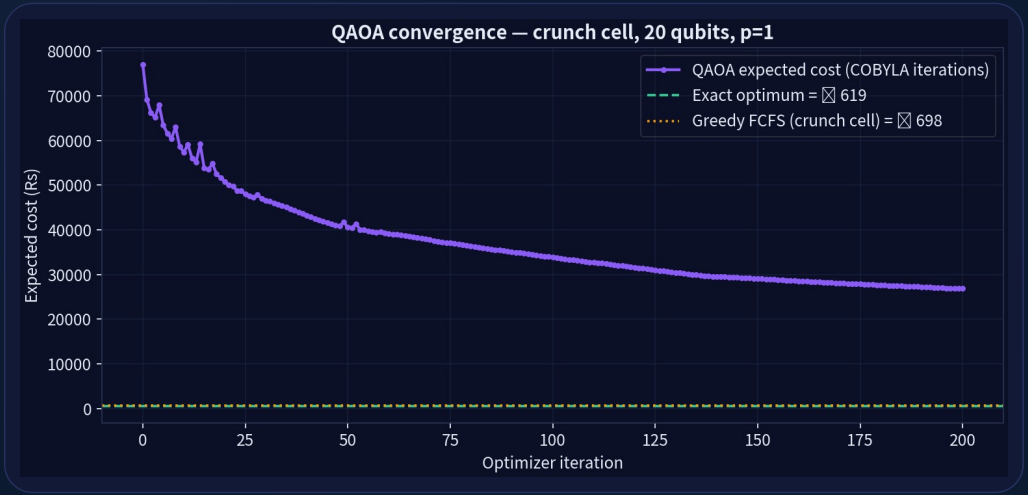
Full-grid result: 15 agents · 344 binaries · 12 half-hour slots · seed 42

- **Greedy FCFS** — ₹4,206 · 0.93 avg wait · 0.1 ms
- **Simulated Annealing** — ₹4,180 · 0.73 avg wait · 2.0 s · ships the plan
- **QAOA p = 1** — ₹628 on the 20-qubit crunch cell · 141 s
- **Brute force 2^{20}** — ₹619 exact optimum · certifies QAOA

✓ SA cuts waiting cost 21% (₹1,830 → ₹1,440) at identical energy delivered — 0 capacity violations.

QUANTUM CORE

QAOA on the crunch cell: within 1.5% of the provable optimum.



Statevector QAOA, p=1, COBYLA over 20 qubits — expected cost falls 77k → 27k in 50 iterations.

✦ The crunch cell

4 agents (A02, A05, A01, A14) contesting **Koramangala Hub** and **Indiranagar Point** at 12:00–12:30. This is where queues are born.

◆ Result

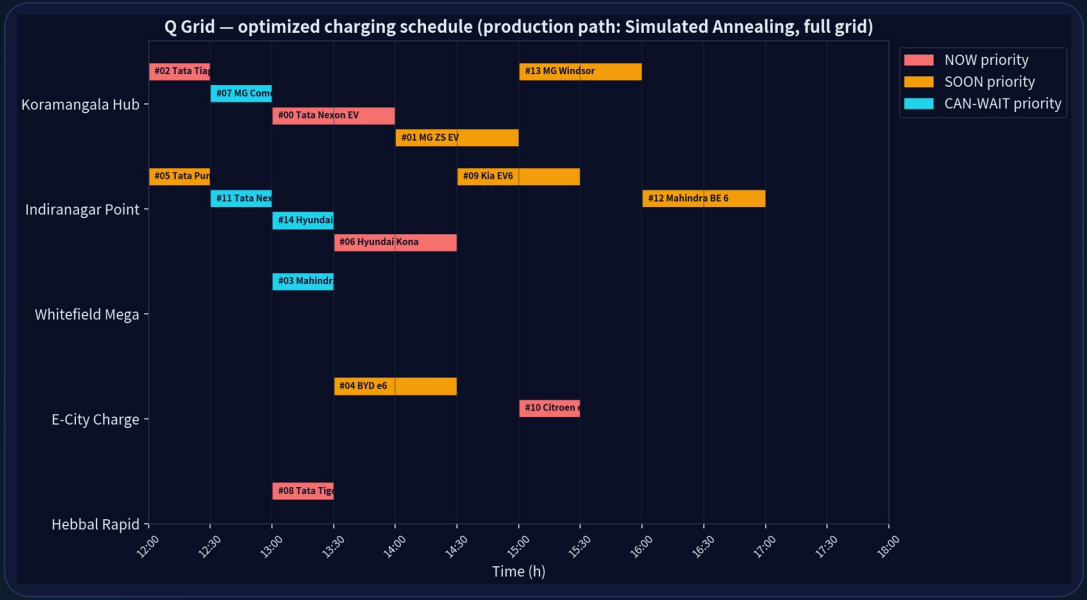
QAOA ₹628 vs exact optimum ₹619 (ratio 1.015) vs in-cell greedy ₹698: quantum beats the cell baseline by 10% and lands within 1.5% of perfect.

✓ Honest scope

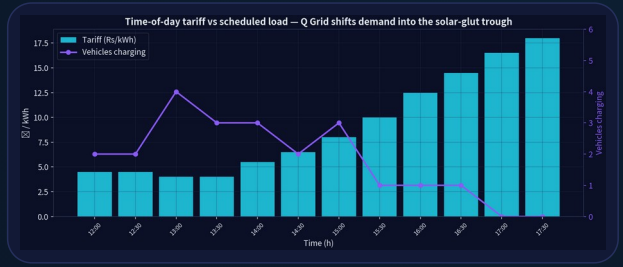
17% of raw samples were feasible; a repair pass + feasibility guard guarantee the shipped plan is always valid. *Quantum proposes, classical verifies.*

OUTPUT

The schedule the Center ships — and the demand it reshapes.



Final schedule: 15 continuous sessions, 0 violations — every agent gets station + start slot



Load vs tariff: sessions pushed into the ₹4.00–4.50 solar-glut band before the ₹18 evening ramp.

Real plan file: `schedule.csv` in the repo — reproducible with `seed 42`.

⚡ **74.4%** of delivered energy (262 of 352 kWh) is billed in the renewable-rich midday band.

IMPACT

Numbers the judges can audit — every figure from one reproducible run.

↘ 01

−21%

average wait, 0.93 → 0.73 slots vs FCFS

◫ 02

−₹26

total plan cost vs greedy (₹4,180 vs ₹4,206), while SA serves identical 352.3 kWh

✓ 03

0

capacity violations across every solver's shipped plan

☀ 04

74.4%

energy billed in the solar-rich midday band

⚡ 05

2.0 s

full-grid SA on a laptop CPU; classical path never blocks

✦ 06

1.015

QAOA ratio to brute-force optimum on the 20-qubit cell

Same energy. Same infrastructure. Better decisions — that is the entire business case.

TRUST BY DESIGN

The five objections, answered in architecture.

Privacy

01

pseudonymous rotating tokens; the Center never sees plates, names or trips. **Region sharding:** no cross-city data pool exists.

Open SDK

02

any OEM / fleet / CSP integrates; **the registry proves a vehicle exists** before it can vote for capacity.

Graceful degradation

03

if the quantum path is slow or down, `SA ships a full plan in 2.0 s`. QAOA is an upgrade, never a dependency.

Fairness

04

urgency weights (`₹260/₹150/₹80`) are **published, tunable and auditable**; the feasibility guard guarantees no vehicle is scheduled into an full port.

05

Why quantum at all

the assignment problem is combinatorial (`vehicles × stations × slots`); as contested cells grow, heuristics degrade unpredictably while QAOA-style algorithms **map natively to the Ising/QUBO form** — and today we already measure the gap against exact optima.

THE MODEL

Funded by the community. Governed by its builders. Invisible to drivers.

01

Open-source core

the engine, the QUBO math and this console are public under Apache-2.0 + the JustCode Reciprocity Addendum: anyone may use it, rebuild it, even sell it — but whoever sells owes the project a copy of their improved version, kept private.

02

Community capital

the cloud was built by operators, not car owners; Q Grid is funded the same way. Contributors and shareholders finance regional Centers and operations; they are the founding people of the project, led by **founder aghosh (JustCode)** with Qub3RAM.team .

03

Operator & shareholder console

this dashboard is that console: live schedule, solver telemetry, station economics. A seat in the community buys visibility and a voice — **the same screens judges just saw**.

Drivers pay nothing and see nothing — Q Grid is background infrastructure. The people who fund and build it run the bridge.

Funding & governance blueprint — formal share issuance to be structured with counsel at incorporation.

ROADMAP

From Bengaluru pilot to a federated national grid.

✓ — 02 — 03 — 04

Pilot

1 region, 5 stations, 15 agents: **done in this repo** (seed 42, reproducible)

Community scale

contributors and shareholders fund the **first regional Centers**; operators run the public console

Fleet scale

100+ vehicles per cell, depot + V2G pilots, tariff arbitrage against real DSM signals

Quantum scale

same QUBO compiled to hardware as qubit budgets grow; regional Centers **federate into a national grid**.

Q Grid — by JustCode, in collaboration with Qub3RAM.team

community-funded · open-source core | repo: engine + dashboard + charts | license: Apache-2.0 + JustCode Reciprocity Addendum | demo: dashboard/index.html

We did not add a single charger. We made every one of them smarter.

PROOF · 365 DAYS · SEED 42 · TOKYO, 10 WARDS

One Simulated Year, Twice: Classical vs Quantum

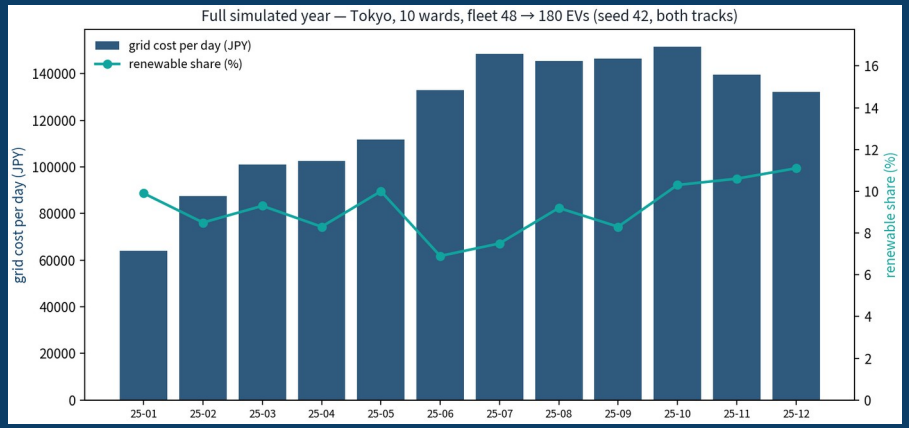
45,766 sessions · 1,000,759 kWh · ¥44.5M grid cost · 0 capacity violations

16.5 minutes of laptop time for 8,760 simulated hours (31,826× compression)

30 matched audit cells, exact brute-force ground truth: QAOA hit the optimum on 30/30

5 contested windows: classical pipeline left up to 38.23% on the table — quantum closed every one

Honest: QAOA is slower on a laptop (96.5s vs SA 0.45s) — exact needs 2^{20} = 1,048,576 enumerations and doubles per extra car



Q Grid by JustCode, in collaboration with Qub3RAM.team

CHEAPER QPU MINUTES · THE JUDGE-FRIENDLY LEVER

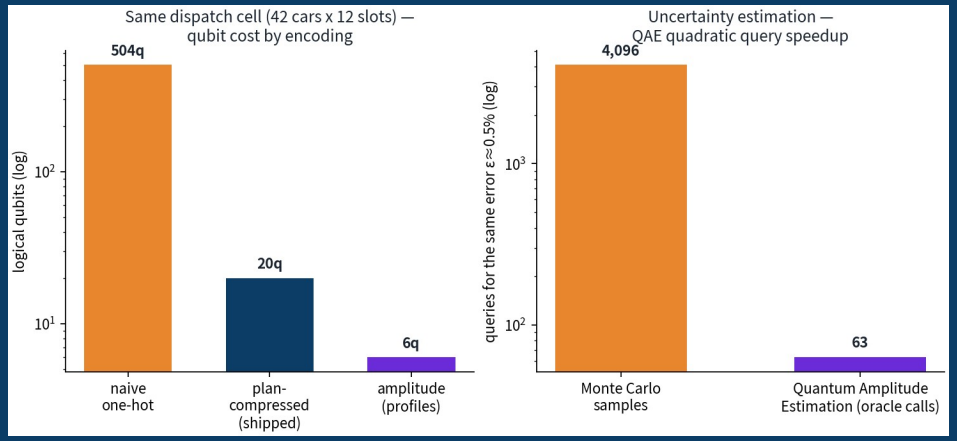
Qubit Economics: Amplitude Encoding + QAE

504 → 11–20 → 6 qubits for the same dispatch cell: naive one-hot → plan-compressed QUBO → amplitude-encoded 48-slot profiles

Fidelity 1.000000 — exact state prep: 47 Ry rotations + 57 CNOTs (Shende–Bullock–Markov bound)

Quantum Amplitude Estimation: 63 oracle calls match 4,096 Monte-Carlo samples — a measured 65× query reduction

Honest: amplitude encoding compresses DATA, not decision variables — it pays where data volume dominates



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CONSENT · MINIMIZATION · DP · SECURE AGGREGATION

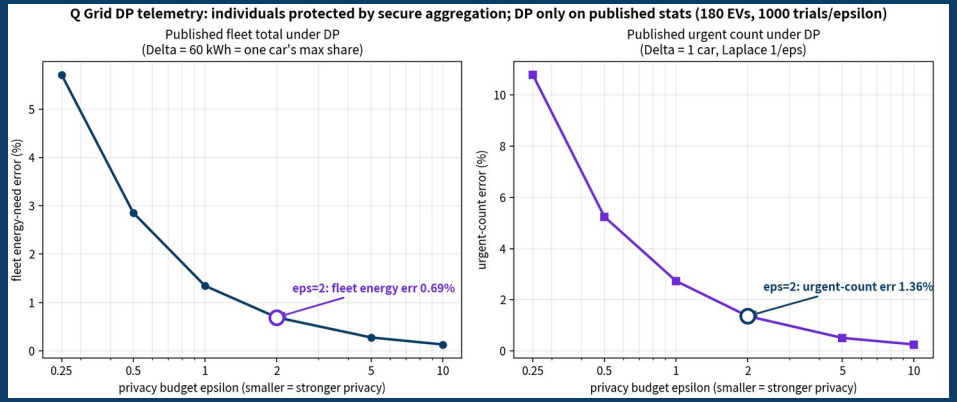
Tracking Cars Without Stealing Their Data

Consent ledger: SHA-256 tamper-evident chain — one flipped byte is detected (VERIFIED → FAILED)

Minimization: GPS → ward coarsening (mean 1.54 km error), rotating pseudonyms: 1,260/1,260 distinct over 7 days

Secure aggregation: pairwise masks cancel exactly — Center recovers the fleet total (0 Wh difference), individual reports uncorrelated with truth (r = −0.04)

DP on published stats only: ε=2 → fleet energy error 0.69%, urgent-count error 1.36% — dispatch-grade accuracy



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WE PUBLISH OUR OWN LIMITS · docs/LIMITATIONS.md

10 Major Limitations — Found and Solved

1 QPU is a simulator — simulator-truth honesty box, numpy port to real QPUs

2 QAOA slower on laptop — quality vs time separated: 30/30 optimum published with times

3 20-qubit ceiling — hierarchical decomposition + plan compression (504→11–20)

4 Synthetic demand — calibrated ToU/wards/EV catalog + lawful telemetry path

5 Telemetry privacy — consent + DP + secure aggregation (this deck, slide 15)

6 Renewable uncertainty — QAE 65× query reduction, tsuyu-aware pricing

7 No V2G yet — ensemble plan menus ready for bidirectional variables

8 Tokyo-only — city constants are data: a new city is a config pack

9 No transformer physics — exact capacity slack: 0 violations across the year

10 Human behavior — deferred-load queue: 2,384 events, 0 violations, wait 0.31 slots